DS 2002 Capstone Project

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Pipeline

Cloud + API Integration

- Authenticated with Google Cloud Storage (GCS)
- Loaded raw data from cloud bucket
- Integrated external weather API (Open-Metro)
- Merged weather data with charging sessions + demand data
- Saved final cleaned dataset to SQLite + GCS

GCS → Data Cleaning → Weather API → Merge → Final Dataset → Upload to GCS

```
charging_weather.head()
charging_weather[["station_id", "date", "temperature_2m_max", "temperature_2m_min", "precipitation_sum"]].head(10)
```

	station_id	date	temperature_2m_max	temperature_2m_min	precipitation_sum
0	STN013	2025-03-16	21.8	10.9	14.7
1	STN012	2025-10-19	24.0	13.1	7.5
2	STN007	2025-12-30	2.2	-3.3	0.0
3	STN006	2025-04-28	25.0	6.8	0.0
4	STN012	2025-09-14	28.3	14.3	0.0
5	STN017	2025-08-06	23.7	18.1	0.5
6	STN016	2025-05-24	22.8	8.7	0.0
7	STN005	2025-09-24	26.3	17.0	8.1
8	STN010	2025-01-16	6.1	-12.6	0.7
9	STN005	2025-03-11	21.6	1.0	0.0

Cleaning

- 2 Cleaning Decisions: Mixed datetime formats and Inconsistent station ID formatting
- General data quality issues: inconsistent formatting, fragmented identifiers, duplicate records, invalid values, categorical inconsistencies, missing data
- Data cleaning changes: removed duplicate records, standardized datetime formats, normalized text fields, standardized IDs into consistent formatting, fixed inconsistent categorical values, converted numeric fields into proper data types, handled missing values, removed invalid values, ensured reliable joining across tables
- General trade-off: Data Retention v. Strict Consistency

#	File	Column(s)	Issue	Rows Affected (approx)
1	raw-data_grid_operators	peak_capacity_mw	"~550" value when using only numeric values	1
2	raw-data_grid_operators	cost_per_kwh	Missing value	1
3	raw-data_vehicle_types	vehicle_id	Mulitple SKUs for same product	43 (all)
4	raw-data_vehicle_types	vehicle_name	Mulitple names for same vehicle	43 (all)
5	raw-data_vehicle_types	vehicle_class	Inconsistent formatting	43 (all)
6	raw-data_vehicle_types	connector_type	Inconsistent formatting	43 (all)
7	raw-data_vehicle_types	battery_kwh	Missing values	15
8	raw-data_vehicle_types	battery_kwh	Inconsistent formatting for numeric value	1
9	raw-data_vehicle_types	manufacturer	Inconsistent formatting	43 (all)
10	raw-data_vehicle_types	manufacturer	Missing value	1
11	raw-data_station_locations	station_id	Duplicate values	2
12	raw-data_station_locations	state	Inconsistent formatting	4
13	raw-data_station_locations	latitude	Inconsistent formatting for decimal count	~11
14	raw-data_station_locations	latitude	Missing value	1
15	raw-data_station_locations	longitude	Missing value	1
16	raw-data_station_locations	longitude	Inconsistent formatting for decimal count	~11
17	raw-data_station_locations	region	Missing value	1
18	raw-data_charging_sessions	station_id	Inconsistent formatting	~13000
19	raw-data_charging_sessions	vehicle_id	Multiple SKUs for same vehicle	~13000
20	raw-data_charging_sessions	session_start	Inconsistent formatting on dates	~10000

Cleaning

Mixed Datetime Formats

- Before: session_start and session_end had multiple inconsistent formats → 2025-03-16 17:06:25, 10-19-2025 17:10:08, 12/30/2025 07:30, 2025/08/06 11:54
- Decision: Implemented a mixed-format datetime parser to standardize timestamps into consistent format →YYYY-MM-DD HH:MM:SS
- After: all timestamp data was preserved and converted into consistent formatting
- Impact: Created intreptable datetime data to aid in analysis and discover key relationships
- Trade-off: data retention & consistent formatting with small risk of incorrectly interprtting date formats

session_start	session_end
2025-03-16 17:06:25	2025-03-16 18:30:25
10-19-2025 17:10:08	10-19-2025 18:40:08
12/30/2025 07:30	12/30/2025 08:21
04-28-2025 02:35:41	04-28-2025 04:52:41
2025-09-14 14:28:19	2025-09-14 16:39:19
2025/08/06 11:54	2025/08/06 12:44
2025-05-24 16:18:09	2025-05-24 17:54:09
2025-09-24 17:56:46	2025-09-24 18:10:46
01/16/2025 20:56	01/16/2025 22:49
2025-03-11 18:01:00	2025-03-11 18:38:00

session_start	session_end
2025-03-16 17:06:25	2025-03-16 18:30:25
2025-10-19 17:10:08	2025-10-19 18:40:08
2025-12-30 07:30:00	2025-12-30 08:21:00
2025-04-28 02:35:41	2025-04-28 04:52:41
2025-09-14 14:28:19	2025-09-14 16:39:19
2025-08-06 11:54:00	2025-08-06 12:44:00
2025-05-24 16:18:09	2025-05-24 17:54:09
2025-09-24 17:56:46	2025-09-24 18:10:46
2025-01-16 20:56:00	2025-01-16 22:49:00
2025-03-11 18:01:00	2025-03-11 18:38:00

Cleaning

Inconsistent Station ID Formatting

- Before: station IDs appeared in inconsistent formats
  → STN16, STN5, STN016
- Decision: Standardized all station IDs to a fixed format
  STN### → STN16 to STN016
- After: All station IDs follow a consistent format
- Impact: Reliable joins across dataset and prevents risk of mismatching values
- Trade-off: Consistency prioritzed to ensure accurate joining but follows assumption that station IDs follow the same naming conventions

station_id
STN-013
STN-012
STN-007
STN-006
STN-012
STN-017
STN16
STN-005
STN-010
STN-005

station_id
STN013
STN012
STN007
STN006
STN012
STN017
STN016
STN005
STN010
STN005

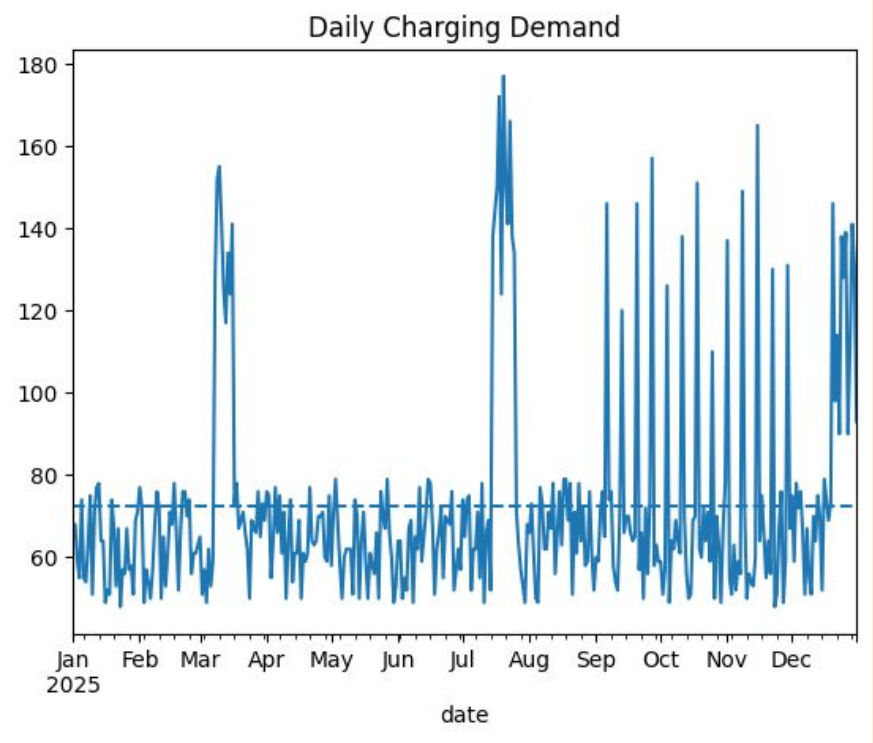
Analysis:
Charging Demand, Weather, and Connector Types

We analyzed EV charging data to understand usage patterns, external factors, and infrastructure needs.

- Q1: Demand surge identification
- Q2: Vehicle consolidation and true charging usage
- Q3: Weather effects on charging demand and grid load
- Q5: Connector type trends and recommendation

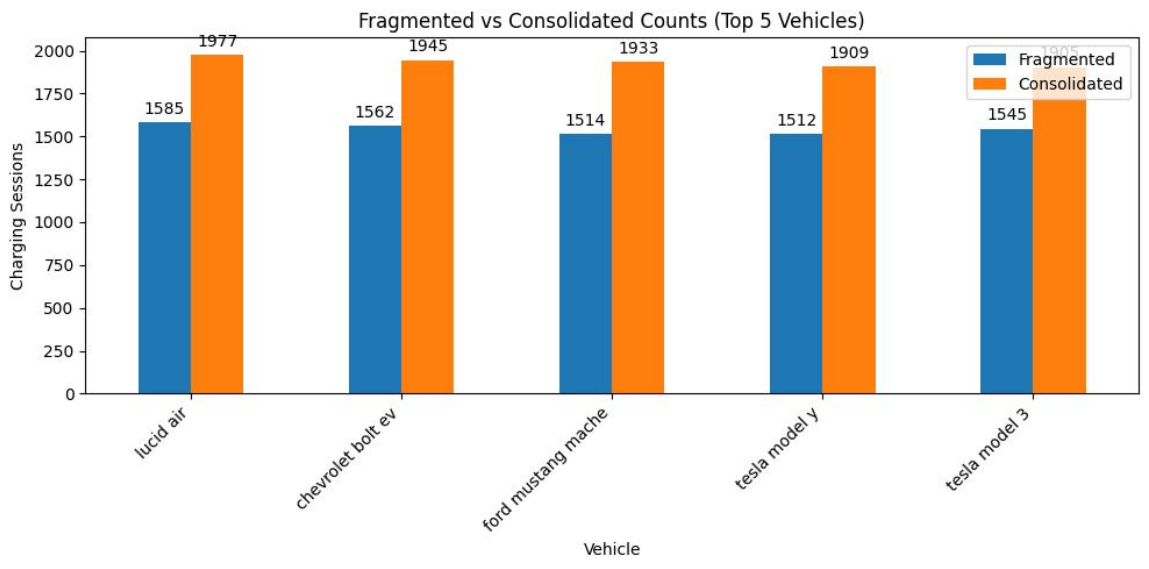
Demand Surge Identification

- Baseline: ~70 sessions per day
- Peak demand occurs in late July–August (~170–180 sessions)
- Repeated spikes in fall (Sep–Dec)



Vehicle Consolidation Problem

- Some vehicles are split across multiple IDs
- “Fragmented” = only highest single ID count
- “Consolidated” = total across all IDs



Differences between fragmented and consolidated counts show that raw data underestimates charging demand for each vehicle.

Impact of Consolidation
What the data shows

- Consolidated counts are significantly higher for every vehicle
- Data cleaning directly changes conclusions
- Each vehicle is undercounted when IDs are split
- Important for accurate demand planning
- Differences are around ~300-500 sessions per vehicle
- Raw data would underestimate true demand

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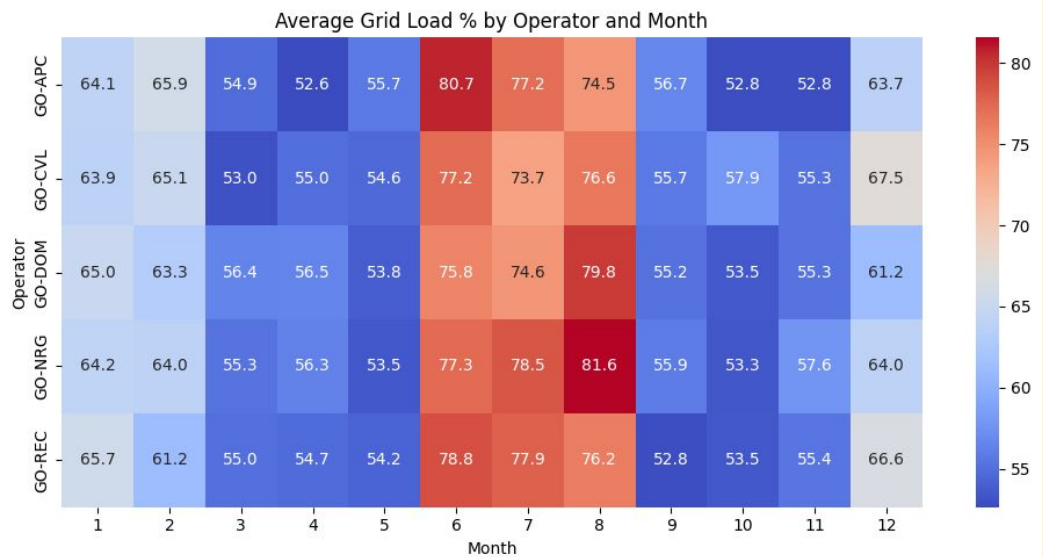
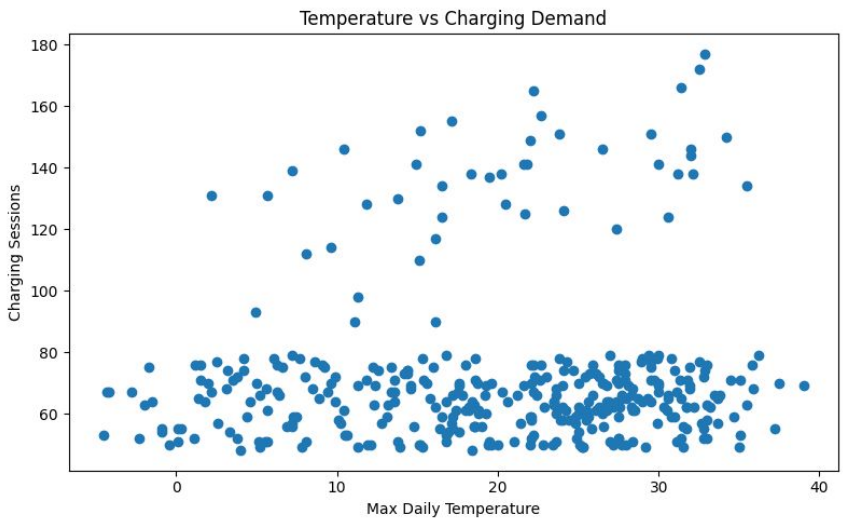
Weather and Charging Demand

Temperature vs. Daily Charging Sessions





Average Grid Load by Month

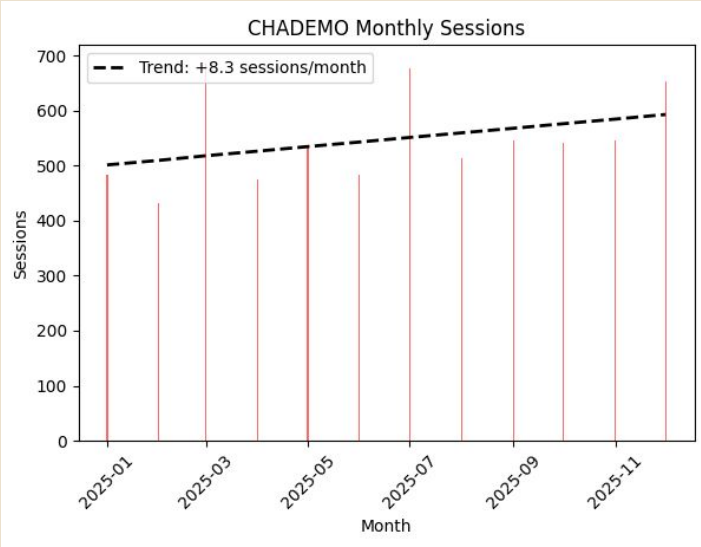
- Weak relationship between temperature and charging
- Grid load varies strongly by month

Key Findings From Weather and Charging Demand

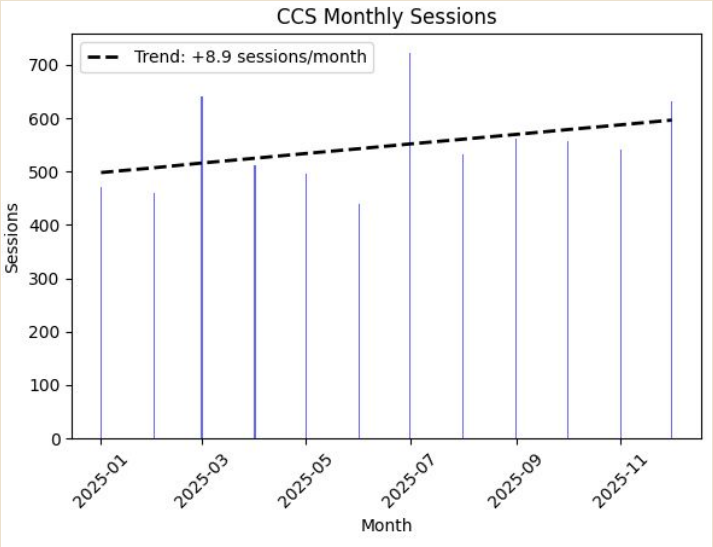
- Temperature does not strongly predict charging demand
- Charging levels stay fairly consistent across temperatures
- Grid load peaks in summer (June - August)
- Seasonal demand has a larger impact than daily temperature

Connector Type Investigation

CHAdeMO



CCS



Connector Type Investigation

Evidence

Based on the actual trend data, CHAdeMO is not declining:

- CHAdeMO slope: +8.33 sessions/month
  - A positive slope, suggesting growth.
- CCS slope: +8.94 sessions/month
  - A positive slope, suggesting growth.

Recommendation

Both connector types are growing at nearly identical rates. This means **the apparent CHAdeMO decline** suggested by the overall session counts is most likely **a data artifact**.

Based on this data, there is **no strong case to cut CHAdeMO funding**. A reasonable approach is to **maintain existing CHAdeMO ports** but direct any new infrastructure investment toward CCS, which has stronger long-term manufacturer support as newer EVs support CCS more.

Thank you!